

Linear Test 6 Math

The SAT[®]

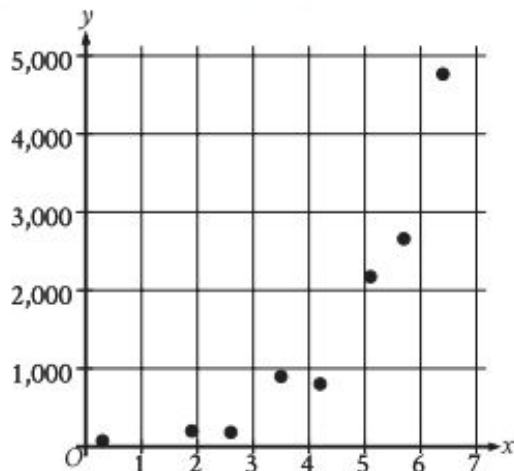
Practice

Test #6

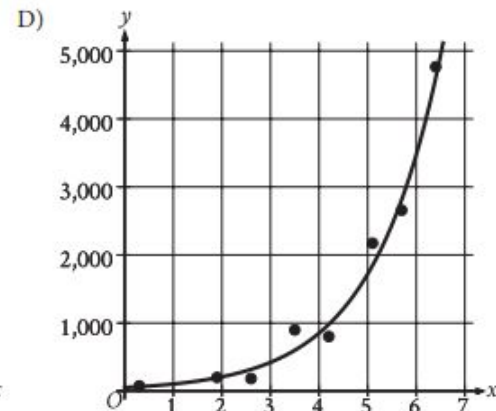
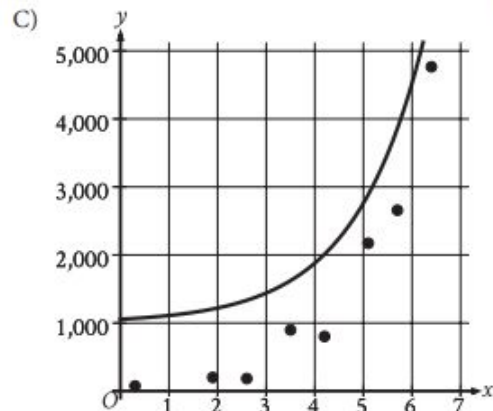
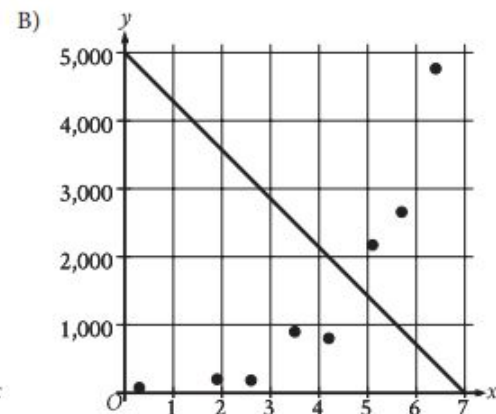
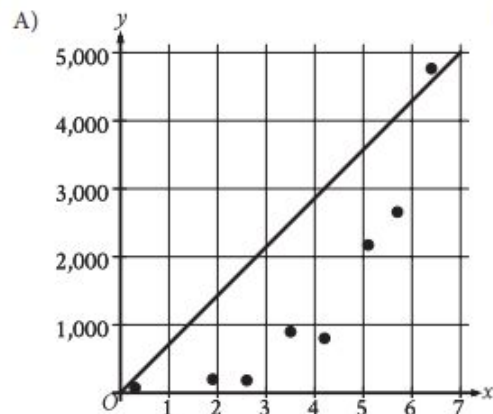


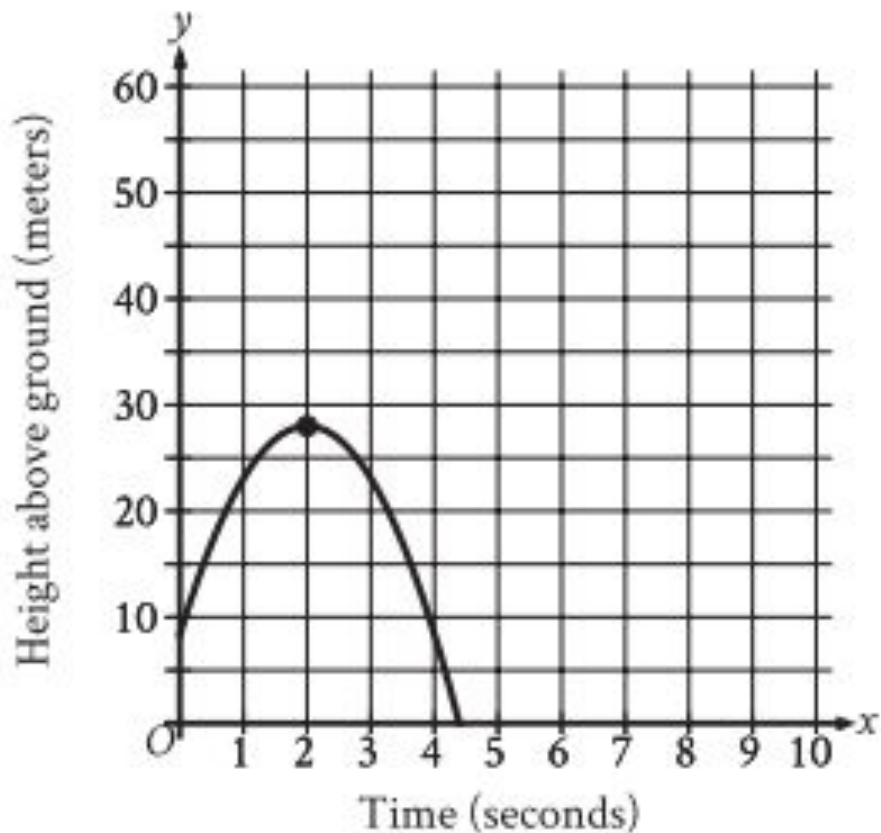
LT6 - M1.2

The scatterplot shows the relationship between two variables, x and y .



Which of the following graphs shows the most appropriate model for the data?





An object was launched upward from a platform. The graph shown models the height above ground, y , in meters, of the object x seconds after it was launched. For which of the following intervals of time was the height of the object increasing for the entire interval?

- A) From $x = 0$ to $x = 2$
- B) From $x = 0$ to $x = 4$
- C) From $x = 2$ to $x = 3$
- D) From $x = 3$ to $x = 4$

How many yards are equivalent to 1,116 inches?
(1 yard = 36 inches)

$$f(x) = x + b$$

For the linear function f , b is a constant.

When $x = 0$, $f(x) = 30$. What is the value of b ?

A) -30

B) $-\frac{1}{30}$

C) $\frac{1}{30}$

D) 30

$$P(t) = 1,800(1.02)^t$$

The function P gives the estimated number of marine mammals in a certain area, where t is the number of years since a study began. What is the best interpretation of $P(0) = 1,800$ in this context?

- A) The estimated number of marine mammals in the area was 102 when the study began.
- B) The estimated number of marine mammals in the area was 1,800 when the study began.
- C) The estimated number of marine mammals in the area increased by 102 each year during the study.
- D) The estimated number of marine mammals in the area increased by 1,800 each year during the study.

LT6 - M1.19

The table shown summarizes the number of employees at each of the 17 restaurants in a town.

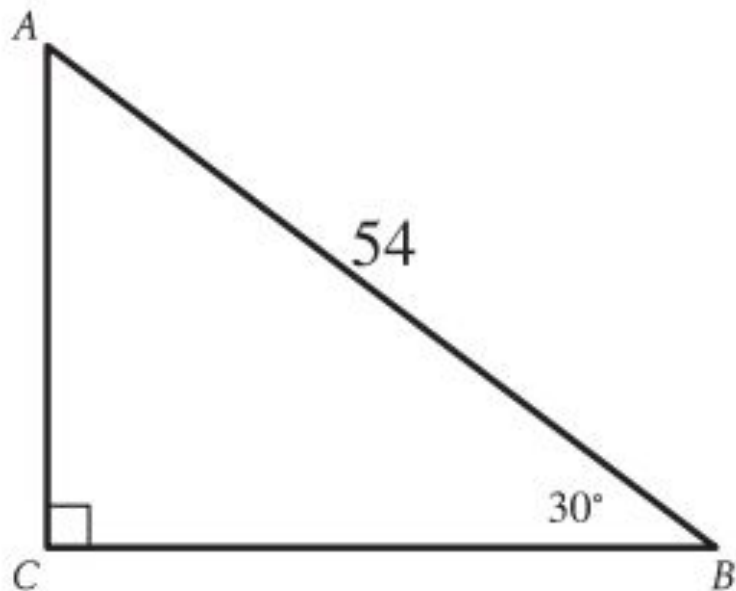
Number of employees	Number of restaurants
2 to 7	2
8 to 13	4
14 to 19	2
20 to 25	7
26 to 31	2

Which of the following could be the median number of employees for the restaurants in this town?

- A) 2
- B) 9
- C) 15
- D) 21

In 2008, Zinah earned 14% more than in 2007, and in 2009 Zinah earned 4% more than in 2008. If Zinah earned y times as much in 2009 as in 2007, what is the value of y ?

- A) 0.5600
- B) 1.0056
- C) 1.1800
- D) 1.1856



Note: Figure not drawn to scale.

Right triangle ABC is shown. What is the value of $\tan A$?

- A) $\frac{\sqrt{3}}{54}$
- B) $\frac{1}{\sqrt{3}}$
- C) $\sqrt{3}$
- D) $27\sqrt{3}$

At the time that an article was first featured on the home page of a news website, there were 40 comments on the article. An exponential model estimates that at the end of each hour after the article was first featured on the home page, the number of comments on the article had increased by 190% of the number of comments on the article at the end of the previous hour. Which of the following equations best represents this model, where C is the estimated number of comments on the article t hours after the article was first featured on the home page and $t \leq 4$?

A) $C = 40(1.19)^t$

B) $C = 40(1.9)^t$

C) $C = 40(19)^t$

D) $C = 40(2.9)^t$

In right triangle ABC , angle C is the right angle and $BC = 162$. Point D on side AB is connected by a line segment with point E on side AC such that line segment DE is parallel to side BC and $CE = 2AE$. What is the length of line segment DE ?

The ratio of the length of line segment XY to the length of line segment ZV is 6 to 1. If the length of line segment XY is 102 inches, what is the length, in inches, of line segment ZV ?

- A) 17
- B) 96
- C) 102
- D) 612

A rectangle has an area of 155 square inches. The length of the rectangle is 4 inches less than 7 times the width of the rectangle. What is the width of the rectangle, in inches?

4, 10, 18, 4, 4, 5, 6, 5

What is the median of the data set shown?

- A) 4
- B) 5
- C) 7
- D) 14

A right circular cylinder has a volume of 432 cubic centimeters. The area of the base of the cylinder is 24 square centimeters. What is the height, in centimeters, of the cylinder?

- A) 18
- B) 24
- C) 216
- D) 10,368

LT6 - M2.18

Line k is defined by $y = 7x + \frac{1}{8}$. Line j is perpendicular to line k in the xy -plane. What is the slope of line j ?

A) -8

B) $-\frac{1}{7}$

C) $\frac{1}{8}$

D) 7

If $4^{8c} = \sqrt[3]{4^7}$, what is the value of c ?

$$48x - 64y = 48y + 24$$

$$ry = \frac{1}{8} - 12x$$

In the given system of equations, r is a constant. If the system has no solution, what is the value of r ?